

# Bagged Offset (production)

offset family, scored against the market on 2015-2026, 5 seed(s), from 7953760a028f (dirty)

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## Model identity and provenance

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

|                                                                              |                                                                |                                                                                 |
|------------------------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------------------------|
| <div>MODEL</div> <div>Bagged Offset<br/>(production)</div> <div>offset</div> | <div>COMPARATOR</div> <div>the market</div>                    | <div>SEEDS</div> <div>5</div> <div>7, 17, 42, 123, 256</div>                    |
| <div>TEST YEARS</div> <div>12</div> <div>2015-2026</div>                     | <div>BOOSTING ROUNDS</div> <div>400</div> <div>3-fold CV</div> | <div>SOURCE COMMIT</div> <div>7953760a028f</div> <div>dirty at generation</div> |

Config digest sha256:cb9351b6601ede16f58a647a4b844d09ee84c3398155cdc2ee48b917081c41a9, model view bagged\_offset, artifacts read from /Users/jlcheung/PycharmProjects/hkjc-mod/results/model\_report.

### The generating source tree was dirty

This run's provenance records uncommitted changes, so the commit named above does not fully describe the code that produced these numbers. Treat the figures as indicative until the report is regenerated from a clean tree.

## ENVIRONMENT

| Component | Version |
|-----------|---------|
| numpy     | 2.5.2   |
| pandas    | 3.0.5   |
| python    | 3.12.11 |
| xgboost   | 3.4.1   |

Deltas in this report use model\_minus\_market\_log\_loss: race\_multinomial\_log\_loss, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

## Executive summary

|                                                                          |                                   |                                          |
|--------------------------------------------------------------------------|-----------------------------------|------------------------------------------|
| <div>MEAN DELTA</div> <div>-0.004647</div> <div>negative is better</div> | <div>RACES</div> <div>8,675</div> | <div>YEARS BETTER</div> <div>11/12</div> |
|--------------------------------------------------------------------------|-----------------------------------|------------------------------------------|

RACE-LEVEL WIN RATE

54.3%

Bagged Offset (production) is better than its comparator by 0.004647 race\_multinomial\_log\_loss, measured against the market on 8,675 races. The direction holds in 11 of 12 evaluated years. Race by race it is ahead 54.3% of the time, which is the more honest figure: a mean delta can be carried by a few races. 339 bets were selected under the staking rules below, on retrospective odds that could not have been bet into.

## Comparator and evaluation population

The market's own implied probabilities on the same races. This is the right comparator for the production model because the model is built as a correction to it: the booster learns a residual on  $\log(\text{market\_prob})$ , so the market is the baseline by construction, not a chosen benchmark.

TEST RACES

8,675

YEARS

12

2015-2026

Splits present: test, train\_apparent, train\_prequential. Only the forward test split is scored above. The others exist to be compared against it -- an apparent-fit number looks far better than a forward one -- and are never pooled with it.

Out-of-fold margin sources: market\_fallback, model.

### One or more years is partial

2026 does not span a full season, so its per-year figures rest on fewer races than the others. Partial years are marked in every table rather than dropped, because dropping them would silently change the evaluation population.

## Overall probability performance

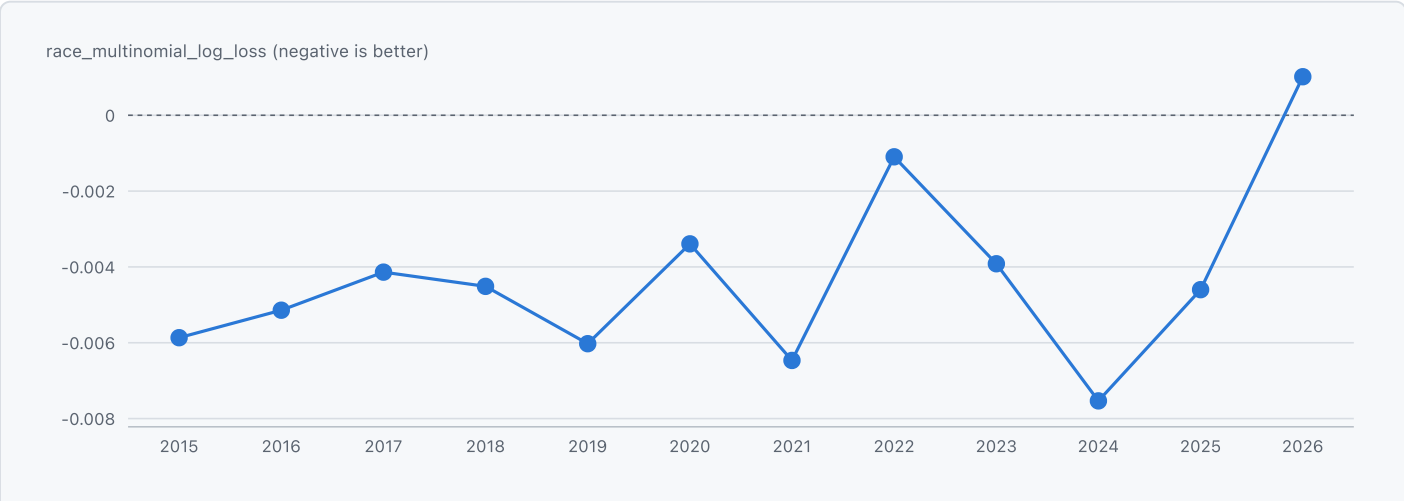
| races | model log loss | market log loss | model minus market log loss | better race share |
|-------|----------------|-----------------|-----------------------------|-------------------|
| 8675  | 2.021454       | 2.026101        | -0.004647                   | 0.543285          |

One row, pooled over every evaluated race. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

## Year-by-year probability performance

| test year | races | model log loss | market log loss | model minus market log loss | better race share | is partial year |
|-----------|-------|----------------|-----------------|-----------------------------|-------------------|-----------------|
| 2015      | 742   | 1.992850       | 1.998720        | -0.005869                   | 0.549865          | no              |
| 2016      | 744   | 2.094135       | 2.099277        | -0.005142                   | 0.564516          | no              |
| 2017      | 772   | 2.017423       | 2.021559        | -0.004136                   | 0.528497          | no              |

| test year | races | model log loss | market log loss | model minus market log loss | better race share | is partial year |
|-----------|-------|----------------|-----------------|-----------------------------|-------------------|-----------------|
| 2018      | 756   | 2.056990       | 2.061504        | -0.004514                   | 0.525132          | no              |
| 2019      | 759   | 1.926801       | 1.932827        | -0.006026                   | 0.548090          | no              |
| 2020      | 799   | 2.007010       | 2.010400        | -0.003390                   | 0.535670          | no              |
| 2021      | 783   | 2.000193       | 2.006659        | -0.006466                   | 0.565773          | no              |
| 2022      | 783   | 2.050100       | 2.051195        | -0.001095                   | 0.519796          | no              |
| 2023      | 796   | 2.020147       | 2.024066        | -0.003919                   | 0.530151          | no              |
| 2024      | 792   | 2.018210       | 2.025744        | -0.007534                   | 0.565657          | no              |
| 2025      | 748   | 2.052795       | 2.057397        | -0.004601                   | 0.554813          | no              |
| 2026      | 201   | 2.027241       | 2.026224        | 0.001016                    | 0.502488          | yes             |



**Mean delta by year** Mean delta for each evaluated year. Zero is the comparator.

▼ Table view

| Category | race_multinomial_log_loss (negative is better) |
|----------|------------------------------------------------|
| 2015     | -0.005869                                      |
| 2016     | -0.005142                                      |
| 2017     | -0.004136                                      |
| 2018     | -0.004514                                      |
| 2019     | -0.006026                                      |
| 2020     | -0.003390                                      |
| 2021     | -0.006466                                      |
| 2022     | -0.001095                                      |
| 2023     | -0.003919                                      |
| 2024     | -0.007534                                      |
| 2025     | -0.004601                                      |
| 2026     | 0.001016                                       |

Ahead of its comparator in 11 of 12 years. 1 row(s) are italicised as partial or fallback years, shown rather than dropped so the evaluation population stays visible.

### Calibration and sharpness

Reliability by pipeline stage, on bins fixed across the whole report so two stages are comparable, pooled from the per-year counts rather than averaged over years. Stages: market, raw\_offset, temperature, market\_blend, bagged.

## MARKET

Pooled over every test year.

| bin       | runners | wins  | predicted probability | observed frequency | observed / expected |
|-----------|---------|-------|-----------------------|--------------------|---------------------|
| 0.00-0.01 | 12,309  | 38    | 0.0063                | 0.0031             | 0.493               |
| 0.01-0.03 | 18,118  | 288   | 0.0171                | 0.0159             | 0.929               |
| 0.03-0.05 | 19,584  | 693   | 0.0365                | 0.0354             | 0.969               |
| 0.05-0.07 | 16,510  | 980   | 0.0625                | 0.0594             | 0.950               |
| 0.07-0.10 | 11,221  | 907   | 0.0874                | 0.0808             | 0.925               |
| 0.10-0.15 | 12,867  | 1,645 | 0.1223                | 0.1278             | 1.045               |
| 0.15-0.20 | 6,866   | 1,226 | 0.1727                | 0.1786             | 1.034               |
| 0.20-0.30 | 6,211   | 1,533 | 0.2418                | 0.2468             | 1.021               |
| 0.30-0.50 | 3,019   | 1,153 | 0.3653                | 0.3819             | 1.046               |
| 0.50-1.00 | 354     | 212   | 0.5542                | 0.5989             | 1.081               |

## RAW\_OFFSET

Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

| bin       | runners | wins  | predicted probability | observed frequency | observed / expected |
|-----------|---------|-------|-----------------------|--------------------|---------------------|
| 0.00-0.01 | 71,707  | 316   | 0.0058                | 0.0044             | 0.759               |
| 0.01-0.03 | 90,687  | 1,502 | 0.0171                | 0.0166             | 0.969               |
| 0.03-0.05 | 100,056 | 3,811 | 0.0368                | 0.0381             | 1.036               |
| 0.05-0.07 | 73,499  | 4,447 | 0.0620                | 0.0605             | 0.976               |
| 0.07-0.10 | 53,920  | 4,526 | 0.0866                | 0.0839             | 0.969               |
| 0.10-0.15 | 60,790  | 7,787 | 0.1224                | 0.1281             | 1.046               |
| 0.15-0.20 | 33,222  | 5,789 | 0.1728                | 0.1743             | 1.008               |
| 0.20-0.30 | 32,301  | 7,735 | 0.2423                | 0.2395             | 0.988               |
| 0.30-0.50 | 16,907  | 6,191 | 0.3692                | 0.3662             | 0.992               |
| 0.50-1.00 | 2,206   | 1,271 | 0.5672                | 0.5762             | 1.016               |

## TEMPERATURE

Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

| bin       | runners | wins  | predicted probability | observed frequency | observed / expected |
|-----------|---------|-------|-----------------------|--------------------|---------------------|
| 0.00-0.01 | 74,865  | 356   | 0.0057                | 0.0048             | 0.836               |
| 0.01-0.03 | 90,629  | 1,557 | 0.0171                | 0.0172             | 1.006               |
| 0.03-0.05 | 99,137  | 3,826 | 0.0367                | 0.0386             | 1.050               |
| 0.05-0.07 | 72,791  | 4,431 | 0.0619                | 0.0609             | 0.983               |
| 0.07-0.10 | 53,074  | 4,485 | 0.0866                | 0.0845             | 0.976               |
| 0.10-0.15 | 59,848  | 7,664 | 0.1224                | 0.1281             | 1.046               |
| 0.15-0.20 | 32,786  | 5,686 | 0.1727                | 0.1734             | 1.004               |
| 0.20-0.30 | 32,279  | 7,710 | 0.2422                | 0.2389             | 0.986               |
| 0.30-0.50 | 17,427  | 6,270 | 0.3697                | 0.3598             | 0.973               |

| bin       | runners | wins  | predicted probability | observed frequency | observed / expected |
|-----------|---------|-------|-----------------------|--------------------|---------------------|
| 0.50-1.00 | 2,459   | 1,390 | 0.5704                | 0.5653             | 0.991               |

## MARKET\_BLEND

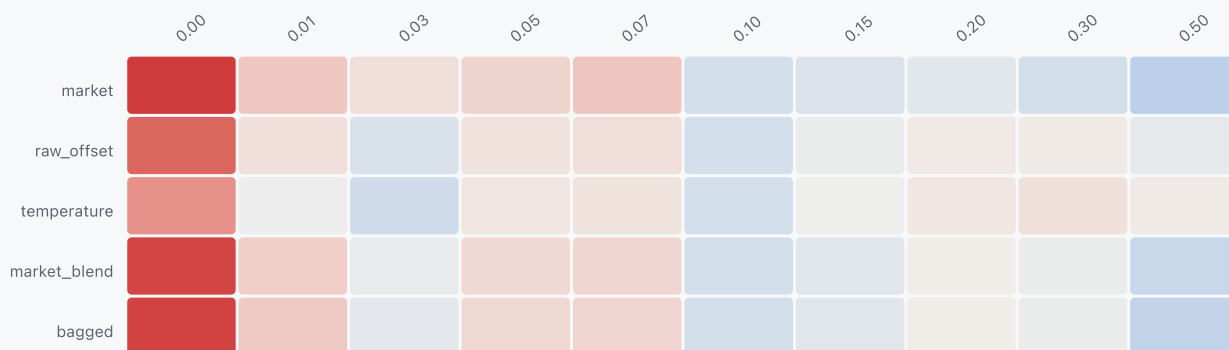
Pooled over every test year and all 5 members (123, 17, 256, 42, 7), so each runner is counted once per member. This is the average member's reliability at this stage, not the bag's.

| bin       | runners | wins  | predicted probability | observed frequency | observed / expected |
|-----------|---------|-------|-----------------------|--------------------|---------------------|
| 0.00-0.01 | 68,725  | 291   | 0.0059                | 0.0042             | 0.712               |
| 0.01-0.03 | 90,588  | 1,457 | 0.0171                | 0.0161             | 0.941               |
| 0.03-0.05 | 99,693  | 3,713 | 0.0368                | 0.0372             | 1.013               |
| 0.05-0.07 | 73,348  | 4,360 | 0.0619                | 0.0594             | 0.960               |
| 0.07-0.10 | 57,417  | 4,731 | 0.0863                | 0.0824             | 0.955               |
| 0.10-0.15 | 61,586  | 7,885 | 0.1224                | 0.1280             | 1.046               |
| 0.15-0.20 | 33,460  | 5,910 | 0.1725                | 0.1766             | 1.024               |
| 0.20-0.30 | 32,197  | 7,753 | 0.2422                | 0.2408             | 0.994               |
| 0.30-0.50 | 16,224  | 6,042 | 0.3691                | 0.3724             | 1.009               |
| 0.50-1.00 | 2,057   | 1,233 | 0.5644                | 0.5994             | 1.062               |

## BAGGED

The bagged probabilities, pooled over every test year.

| bin       | runners | wins  | predicted probability | observed frequency | observed / expected |
|-----------|---------|-------|-----------------------|--------------------|---------------------|
| 0.00-0.01 | 13,735  | 58    | 0.0059                | 0.0042             | 0.710               |
| 0.01-0.03 | 18,144  | 290   | 0.0171                | 0.0160             | 0.935               |
| 0.03-0.05 | 19,921  | 746   | 0.0368                | 0.0374             | 1.018               |
| 0.05-0.07 | 14,672  | 871   | 0.0619                | 0.0594             | 0.959               |
| 0.07-0.10 | 11,473  | 942   | 0.0863                | 0.0821             | 0.952               |
| 0.10-0.15 | 12,315  | 1,575 | 0.1223                | 0.1279             | 1.045               |
| 0.15-0.20 | 6,700   | 1,184 | 0.1725                | 0.1767             | 1.025               |
| 0.20-0.30 | 6,440   | 1,550 | 0.2422                | 0.2407             | 0.994               |
| 0.30-0.50 | 3,249   | 1,211 | 0.3690                | 0.3727             | 1.010               |
| 0.50-1.00 | 410     | 248   | 0.5646                | 0.6049             | 1.071               |



**Observed over expected by stage and bin** Observed wins over expected wins, on fixed report-wide bins, pooled over years rather than averaged across them. Blue is over-delivery, red under. Colour saturates at +0.30 from 1.0, so a cell further out than that is drawn at the endpoint and its value must be read from the tables above.

The largest departure is the 0.000 to 0.010 bin, and every stage misses there: market 0.49, raw\_offset 0.76, temperature 0.84, market\_blend 0.71, bagged 0.71. market is furthest from 1.0 at 0.49, meaning that bin won fewer races than it priced; temperature is closest at 0.84, so this bin is where the pipeline moves calibration most. On the market stage that bin holds 12,309 runners, the 6th smallest of 10 bins, so the departure is not a thin-bin artefact. Calibration in this bin does not improve monotonically through the pipeline: temperature reaches 0.84 and market\_blend falls back to 0.71, so that step gives back part of the correction rather than adding to it.

## SHARPNESS

| stage        | races | mean entropy | mean normalised entropy | mean top probability | mean field size |
|--------------|-------|--------------|-------------------------|----------------------|-----------------|
| market       | 8,675 | 2.0683       | 0.8263                  | 0.2913               | 12.3411         |
| raw_offset   | 8,675 | 2.0296       | 0.8109                  | 0.3044               | 12.3411         |
| temperature  | 8,675 | 2.0181       | 0.8063                  | 0.3084               | 12.3411         |
| market_blend | 8,675 | 2.0415       | 0.8156                  | 0.3005               | 12.3411         |
| bagged       | 8,675 | 2.0416       | 0.8156                  | 0.3005               | 12.3411         |

Entropy normalised by field size, so an eight-runner race and a fourteen-runner race are on one scale, and averaged over years weighted by races rather than as a mean of yearly means. A sharper distribution is not automatically better: it is only better if the calibration above holds.

## Top-choice movement

| test year | n races | n disagreement races | disagreement share | model top win rate | market top win rate | disagreement mean delta | agreement mean delta | seed top choice agreement |
|-----------|---------|----------------------|--------------------|--------------------|---------------------|-------------------------|----------------------|---------------------------|
| 2015      | 742     | 46                   | 0.061995           | 0.239130           | 0.260870            | -0.015097               | -0.005260            | 0.839130                  |
| 2016      | 744     | 52                   | 0.069892           | 0.250000           | 0.250000            | -0.013700               | -0.004499            | 0.965385                  |
| 2017      | 772     | 56                   | 0.072539           | 0.321429           | 0.107143            | -0.017866               | -0.003062            | 0.871429                  |
| 2018      | 756     | 60                   | 0.079365           | 0.183333           | 0.050000            | -0.001932               | -0.004737            | 0.930000                  |
| 2019      | 759     | 50                   | 0.065876           | 0.300000           | 0.220000            | -0.021397               | -0.004942            | 0.928000                  |
| 2020      | 799     | 43                   | 0.053817           | 0.209302           | 0.186047            | 0.003582                | -0.003787            | 0.902326                  |
| 2021      | 783     | 70                   | 0.089400           | 0.242857           | 0.257143            | -0.017521               | -0.005381            | 0.931429                  |
| 2022      | 783     | 50                   | 0.063857           | 0.240000           | 0.220000            | -0.006338               | -0.000737            | 0.892000                  |
| 2023      | 796     | 45                   | 0.056533           | 0.177778           | 0.222222            | -0.008206               | -0.003662            | 0.924444                  |

| test year | n races | n disagreement races | disagreement share | model top win rate | market top win rate | disagreement mean delta | agreement mean delta | seed top choice agreement |
|-----------|---------|----------------------|--------------------|--------------------|---------------------|-------------------------|----------------------|---------------------------|
| 2024      | 792     | 50                   | 0.063131           | 0.160000           | 0.260000            | 0.004092                | -0.008317            | 0.892000                  |
| 2025      | 748     | 59                   | 0.078877           | 0.203390           | 0.169492            | -0.006463               | -0.004442            | 0.911864                  |
| 2026      | 201     | 11                   | 0.054726           | 0.090909           | 0.363636            | 0.006035                | 0.000726             | 0.909091                  |

How often this model's first pick differs from the market's, and whether it is right when it does. Disagreeing more is not better on its own -- the useful figure is the win rate *conditional* on disagreeing, because a model that disagrees often and loses is worse than one that never disagrees. The last column is how often the seeds pick the same runner as each other, which bounds how much of any disagreement is signal rather than seed noise.

## Ensemble stability

|                 |       |              |
|-----------------|-------|--------------|
| RUNNERS COVERED | YEARS | SEEDS BAGGED |
| 107,059         | 12    | 5            |

Member dispersion per runner. The bag is only worth its cost if members genuinely disagree: an ensemble whose members agree everywhere is one fit with five times the runtime. The per-runner spread across members is in `ensemble_diagnostics.parquet`, and the representative-races section below shows the races where the members disagreed most.

## Feature and correction effects

**These are corrections, not drivers of winning**

The booster learns a correction to  $\log(\text{Implied\_Prob})$ , so a feature's attributed effect is its disagreement with the market -- not its contribution to a horse winning. Read the naive way, the sign inverts.

**1 feature cannot be tested this way.**

`market_rank` belongs to no group, so no ablation can remove it and this method cannot measure its held-out value.

| label            | feature                | mean abs correction | sign agreement | rank stability | seed variation | psi      | practical support | feature type | n years |
|------------------|------------------------|---------------------|----------------|----------------|----------------|----------|-------------------|--------------|---------|
| stable_nonlinear | bt_finish_ewm          | 0.047216            | 0.991667       | 0.775991       | 0.058443       | 0.046310 | 12                | continuous   | 12      |
| stable_positive  | early_style_ewm        | 0.036186            | 0.983333       | 0.570197       | 0.075225       | 0.007011 | 12                | continuous   | 12      |
| stable_nonlinear | market_rank            | 0.029815            | 1              | 0.461308       | 0.104674       | 0.001269 | 12                | continuous   | 12      |
| stable_nonlinear | bt_days_since          | 0.029458            | 0.833333       | 0.248826       | 0.100496       | 0.093612 | 12                | continuous   | 12      |
| stable_positive  | trainer_track_residual | 0.026494            | 0.950000       | 0.241158       | 0.080158       | 0.129261 | 12                | continuous   | 12      |
| stable_nonlinear | draw_pct               | 0.024172            | 0.925000       | 0.373624       | 0.120329       | 0.000533 | 12                | continuous   | 12      |
| stable_nonlinear | class_change           | 0.021257            | 0.766667       | 0.359226       | 0.266349       | 0.006694 | 12                | categorical  | 12      |
| stable_nonlinear | body_wt_change_z       | 0.019413            | 0.950000       | 0.238410       | 0.131227       | 0.010553 | 12                | continuous   | 12      |



| label                | feature         | mean abs<br>correction | sign<br>agreement | rank<br>stability | seed<br>variation | psi      | practical<br>support | feature<br>type | n<br>years |
|----------------------|-----------------|------------------------|-------------------|-------------------|-------------------|----------|----------------------|-----------------|------------|
| stable_nonlinear     | layoff_log      | 0.018617               | 0.941667          | 0.322228          | 0.114803          | 0.103384 | 12                   | continuous      | 12         |
| stable_nonlinear     | rating_z        | 0.017888               | 0.925000          | 0.289509          | 0.249088          | 0.021924 | 12                   | continuous      | 12         |
| stable_nonlinear     | distance_change | 0.016749               | 0.958333          | 0.193844          | 0.167273          | 0.019978 | 12                   | continuous      | 12         |
| stable_nonlinear     | rating_change   | 0.016323               | 0.950000          | 0.313155          | 0.193168          | 0.045097 | 12                   | continuous      | 12         |
| stable_nonlinear     | wt_z            | 0.013995               | 0.958333          | 0.272166          | 0.161691          | 0.035055 | 12                   | continuous      | 12         |
| distribution_shifted | jockey_residual | 0.013914               | 0.833333          | 0.227441          | 0.126905          | 0.794525 | 12                   | continuous      | 12         |
| stable_nonlinear     | pace_pressure   | 0.013054               | 0.885417          | 0.193571          | 0.144272          | 0.064975 | 12                   | continuous      | 12         |
| stable_nonlinear     | trip_excuse_ewm | 0.012064               | 0.796296          | 0.268475          | 0.171276          | 0.058036 | 12                   | continuous      | 12         |
| stable_nonlinear     | setup_weight_z  | 0.011772               | 0.833333          | 0.446406          | 0.272290          | 0.002993 | 12                   | continuous      | 12         |
| stable_nonlinear     | bt_last_behind  | 0.010384               | 0.722222          | 0.200916          | 0.198664          | 0.026302 | 12                   | continuous      | 12         |
| regime_dependent     | closer_setup    | 0.005685               | 0.733333          | 0.775991          | 0.210115          | 0.057756 | 12                   | continuous      | 12         |
| stable_nonlinear     | distance_log    | 0.003923               | 1                 | 0.616682          | 0.190325          | 0.072032 | 12                   | categorical     | 12         |
| regime_dependent     | gear_change     | 0.003599               | 0.583333          | 0.258334          | 0.175974          | 0.018293 | 6                    | binary          | 12         |
| stable_nonlinear     | field_size      | 0.002224               | 0.833333          | 0.479933          | 0.288189          | 0.106798 | 12                   | categorical     | 12         |
| stable_nonlinear     | race_class      | 0.002183               | 1                 | 0.464102          | 0.334889          | 0.055002 | 12                   | categorical     | 12         |
| stable_nonlinear     | course_variant  | 0.001801               | 1                 | 0.395644          | 0.125393          | 0.008740 | 12                   | categorical     | 12         |
| distribution_shifted | going_ord       | 0.001708               | 0.591667          | 0.344006          | 0.451781          | 0.355145 | 12                   | categorical     | 12         |
| stable_negative      | Track_Turf      | 0.001099               | 1                 | 0.500951          | 0.384960          | 0.004738 | 12                   | binary          | 12         |
| stable_negative      | Racecourse      | 0.001089               | 1                 | 0.516146          | 0.108297          | 0.002257 | 12                   | binary          | 12         |

## GROUP ABLATION

| group             | mean delta vs full | bootstrap lower | bootstrap upper | n years | n seeds |
|-------------------|--------------------|-----------------|-----------------|---------|---------|
| barrier_trial     | 0.000956           | 0.000279        | 0.001988        | 12      | 5       |
| body_weight       | 0.000415           | 0.000178        | 0.000844        | 12      | 5       |
| historical_market | 0.000400           | -0.000055       | 0.000895        | 12      | 5       |
| pace              | 0.000637           | -0.000003       | 0.001360        | 12      | 5       |
| preparation       | 0.000461           | -0.000102       | 0.001015        | 12      | 5       |
| race_card         | 0.000598           | 0.000150        | 0.001404        | 12      | 5       |
| race_text         | -0.000126          | -0.000402       | 0.000083        | 12      | 5       |

Each group removed from the full model and refitted, scored held out. The deltas do *not* sum to the whole: features share information, so attributing a total by adding these up overstates it. An interval spanning zero means the group's removal was not distinguishable from noise.

## Representative races

Cohorts chosen deterministically from the artifact rather than picked by hand. A hand-picked race is an illustration; a cohort-selected one is evidence. Each cohort shows one race, with every runner's probability at each stage of the pipeline.

## LARGEST ABSOLUTE MARKET DISAGREEMENT

Race 2020122604, 2020, 14 runners.

| horse number | market prob | raw corrected prob | temperature prob | bagged prob | member min | member max | member std | estimated edge | actual result |
|--------------|-------------|--------------------|------------------|-------------|------------|------------|------------|----------------|---------------|
| 1            | 0.009451    | 0.005767           | 0.005313         | 0.006772    | 0.006487   | 0.007008   | 0.000197   | n/a            | lost          |
| 2            | 0.026524    | 0.019456           | 0.018420         | 0.021010    | 0.020487   | 0.021335   | 0.000341   | n/a            | lost          |
| 3            | 0.074749    | 0.059826           | 0.058088         | 0.063366    | 0.062290   | 0.065016   | 0.001113   | n/a            | lost          |
| 4            | 0.010678    | 0.005492           | 0.005054         | 0.006983    | 0.006765   | 0.007060   | 0.000123   | n/a            | lost          |
| 5            | 0.006374    | 0.005085           | 0.004672         | 0.005203    | 0.005050   | 0.005337   | 0.000109   | n/a            | lost          |
| 6            | 0.003329    | 0.002294           | 0.002070         | 0.002499    | 0.002436   | 0.002551   | 0.000047   | n/a            | lost          |
| 7            | 0.124582    | 0.107552           | 0.105811         | 0.114270    | 0.112366   | 0.116594   | 0.002089   | n/a            | lost          |
| 8            | 0.045680    | 0.031262           | 0.029914         | 0.035285    | 0.034836   | 0.035846   | 0.000380   | n/a            | lost          |
| 9            | 0.456799    | 0.591810           | 0.604948         | 0.552438    | 0.546019   | 0.558495   | 0.004676   | -0.005611      | won           |
| 10           | 0.048367    | 0.032048           | 0.030684         | 0.036719    | 0.036043   | 0.037292   | 0.000449   | n/a            | lost          |
| 11           | 0.120917    | 0.089801           | 0.087990         | 0.099548    | 0.098822   | 0.100531   | 0.000621   | n/a            | lost          |
| 12           | 0.029366    | 0.023954           | 0.022785         | 0.024846    | 0.024517   | 0.025183   | 0.000278   | n/a            | lost          |
| 13           | 0.004031    | 0.002815           | 0.002552         | 0.003047    | 0.002970   | 0.003102   | 0.000054   | n/a            | lost          |
| 14           | 0.039154    | 0.022838           | 0.021700         | 0.028013    | 0.027403   | 0.028731   | 0.000548   | n/a            | lost          |

## MEDIAN MARKET DISAGREEMENT

Race 2018040801, 2018, 14 runners.

| horse number | market prob | raw corrected prob | temperature prob | bagged prob | member min | member max | member std | estimated edge | actual result |
|--------------|-------------|--------------------|------------------|-------------|------------|------------|------------|----------------|---------------|
| 1            | 0.011930    | 0.009787           | 0.009565         | 0.010442    | 0.010261   | 0.010736   | 0.000185   | n/a            | lost          |
| 2            | 0.042696    | 0.037399           | 0.037007         | 0.038480    | 0.037644   | 0.039825   | 0.000845   | n/a            | lost          |
| 3            | 0.057945    | 0.054981           | 0.054598         | 0.054733    | 0.053195   | 0.056256   | 0.001184   | n/a            | lost          |
| 4            | 0.073748    | 0.062108           | 0.061746         | 0.066249    | 0.065487   | 0.067691   | 0.000906   | n/a            | lost          |
| 5            | 0.018437    | 0.017043           | 0.016743         | 0.017303    | 0.016868   | 0.017632   | 0.000359   | n/a            | lost          |
| 6            | 0.067602    | 0.070147           | 0.069816         | 0.068486    | 0.067355   | 0.070001   | 0.001027   | n/a            | lost          |
| 7            | 0.193150    | 0.217565           | 0.218813         | 0.210357    | 0.206102   | 0.214286   | 0.003348   | n/a            | won           |
| 8            | 0.253509    | 0.258221           | 0.260113         | 0.258901    | 0.256060   | 0.263651   | 0.002990   | n/a            | lost          |
| 9            | 0.082779    | 0.076056           | 0.075754         | 0.080244    | 0.079066   | 0.081431   | 0.001085   | n/a            | lost          |
| 10           | 0.021925    | 0.018129           | 0.017819         | 0.019678    | 0.019340   | 0.020155   | 0.000328   | n/a            | lost          |
| 11           | 0.005167    | 0.003249           | 0.003143         | 0.004066    | 0.004013   | 0.004146   | 0.000050   | n/a            | lost          |
| 12           | 0.036874    | 0.035090           | 0.034702         | 0.034833    | 0.034206   | 0.035778   | 0.000582   | n/a            | lost          |
| 13           | 0.124805    | 0.132394           | 0.132544         | 0.127876    | 0.126514   | 0.129880   | 0.001411   | n/a            | lost          |
| 14           | 0.009433    | 0.007830           | 0.007636         | 0.008351    | 0.008216   | 0.008526   | 0.000126   | n/a            | lost          |

## NEAR ZERO MARKET DISAGREEMENT

Race 2015010102, 2015, 7 runners.

| horse number | market prob | raw corrected prob | temperature prob | bagged prob | member min | member max | member std | estimated edge | actual result |
|--------------|-------------|--------------------|------------------|-------------|------------|------------|------------|----------------|---------------|
| 1            | 0.073896    | 0.076165           | 0.075237         | 0.074802    | 0.073220   | 0.077638   | 0.001676   | n/a            | lost          |
| 2            | 0.073896    | 0.077734           | 0.076808         | 0.073347    | 0.071862   | 0.075118   | 0.001314   | n/a            | lost          |

| horse number | market prob | raw corrected prob | temperature prob | bagged prob | member min | member max | member std | estimated edge | actual result |
|--------------|-------------|--------------------|------------------|-------------|------------|------------|------------|----------------|---------------|
| 3            | 0.086474    | 0.093058           | 0.092175         | 0.088951    | 0.087428   | 0.090608   | 0.001203   | n/a            | lost          |
| 4            | 0.225793    | 0.222524           | 0.223039         | 0.224341    | 0.223184   | 0.226068   | 0.001111   | n/a            | won           |
| 5            | 0.088354    | 0.097784           | 0.096922         | 0.091239    | 0.090326   | 0.092606   | 0.001012   | n/a            | lost          |
| 6            | 0.045159    | 0.041376           | 0.040534         | 0.043006    | 0.042392   | 0.043394   | 0.000378   | n/a            | lost          |
| 8            | 0.406428    | 0.391359           | 0.395285         | 0.404314    | 0.400962   | 0.410066   | 0.003596   | n/a            | lost          |

## LARGEST MEMBER DISAGREEMENT

Race 2018040809, 2018, 14 runners.

| horse number | market prob | raw corrected prob | temperature prob | bagged prob | member min | member max | member std | estimated edge | actual result |
|--------------|-------------|--------------------|------------------|-------------|------------|------------|------------|----------------|---------------|
| 1            | 0.085019    | 0.072355           | 0.071790         | 0.079738    | 0.078343   | 0.081416   | 0.001106   | n/a            | lost          |
| 2            | 0.010739    | 0.010026           | 0.009768         | 0.010319    | 0.010205   | 0.010533   | 0.000130   | n/a            | lost          |
| 3            | 0.029149    | 0.030508           | 0.030030         | 0.029715    | 0.029300   | 0.030528   | 0.000475   | n/a            | lost          |
| 4            | 0.272062    | 0.310587           | 0.312337         | 0.303324    | 0.292388   | 0.310721   | 0.007708   | n/a            | lost          |
| 5            | 0.022059    | 0.016174           | 0.015827         | 0.018655    | 0.018421   | 0.018914   | 0.000201   | n/a            | won           |
| 6            | 0.011829    | 0.009327           | 0.009081         | 0.010439    | 0.010171   | 0.010620   | 0.000165   | n/a            | lost          |
| 7            | 0.354863    | 0.367179           | 0.369819         | 0.349041    | 0.336882   | 0.362411   | 0.010483   | n/a            | lost          |
| 8            | 0.003962    | 0.002968           | 0.002859         | 0.003396    | 0.003360   | 0.003448   | 0.000035   | n/a            | lost          |
| 9            | 0.074199    | 0.068215           | 0.067646         | 0.071484    | 0.070032   | 0.073022   | 0.001240   | n/a            | lost          |
| 10           | 0.024733    | 0.017719           | 0.017354         | 0.020747    | 0.020477   | 0.021009   | 0.000255   | n/a            | lost          |
| 11           | 0.005007    | 0.003306           | 0.003188         | 0.004051    | 0.003994   | 0.004089   | 0.000038   | n/a            | lost          |
| 12           | 0.034008    | 0.031472           | 0.030987         | 0.032709    | 0.032180   | 0.033211   | 0.000403   | n/a            | lost          |
| 13           | 0.058299    | 0.047780           | 0.047226         | 0.053267    | 0.051793   | 0.054551   | 0.001071   | n/a            | lost          |
| 14           | 0.014072    | 0.012384           | 0.012088         | 0.013113    | 0.012751   | 0.013368   | 0.000235   | n/a            | lost          |

## LARGEST CORRECT CORRECTION

Race 2017110504, 2017, 12 runners.

| horse number | market prob | raw corrected prob | temperature prob | bagged prob | member min | member max | member std | estimated edge | actual result |
|--------------|-------------|--------------------|------------------|-------------|------------|------------|------------|----------------|---------------|
| 1            | 0.074039    | 0.049241           | 0.047762         | 0.060124    | 0.059456   | 0.060840   | 0.000626   | n/a            | lost          |
| 2            | 0.037019    | 0.022997           | 0.022019         | 0.029146    | 0.028736   | 0.029759   | 0.000401   | n/a            | lost          |
| 3            | 0.110058    | 0.062337           | 0.060708         | 0.084594    | 0.083372   | 0.086696   | 0.001246   | n/a            | lost          |
| 4            | 0.019864    | 0.012757           | 0.012093         | 0.015641    | 0.015388   | 0.016045   | 0.000265   | n/a            | lost          |
| 5            | 0.542952    | 0.687631           | 0.697607         | 0.622930    | 0.617359   | 0.626581   | 0.003445   | n/a            | won           |
| 6            | 0.018510    | 0.011425           | 0.010810         | 0.014199    | 0.013875   | 0.014578   | 0.000296   | n/a            | lost          |
| 7            | 0.107162    | 0.095625           | 0.093806         | 0.101747    | 0.099940   | 0.104834   | 0.001904   | n/a            | lost          |
| 8            | 0.012927    | 0.009687           | 0.009140         | 0.010816    | 0.010541   | 0.011001   | 0.000194   | n/a            | lost          |
| 9            | 0.008950    | 0.006101           | 0.005711         | 0.007118    | 0.006914   | 0.007263   | 0.000140   | n/a            | lost          |
| 10           | 0.010859    | 0.008178           | 0.007694         | 0.009178    | 0.008981   | 0.009515   | 0.000210   | n/a            | lost          |
| 11           | 0.003365    | 0.002395           | 0.002207         | 0.002706    | 0.002624   | 0.002773   | 0.000056   | n/a            | lost          |

| horse number | market prob | raw corrected prob | temperature prob | bagged prob | member min | member max | member std | estimated edge | actual result |
|--------------|-------------|--------------------|------------------|-------------|------------|------------|------------|----------------|---------------|
| 12           | 0.054295    | 0.031624           | 0.030444         | 0.041801    | 0.041229   | 0.042602   | 0.000586   | n/a            | lost          |

## LARGEST INCORRECT CORRECTION

Race 2019022008, 2019, 12 runners.

| horse number | market prob | raw corrected prob | temperature prob | bagged prob | member min | member max | member std | estimated edge | actual result |
|--------------|-------------|--------------------|------------------|-------------|------------|------------|------------|----------------|---------------|
| 1            | 0.067221    | 0.072843           | 0.072361         | 0.069313    | 0.067374   | 0.070584   | 0.001381   | n/a            | lost          |
| 2            | 0.016133    | 0.017360           | 0.016907         | 0.016708    | 0.016510   | 0.017090   | 0.000225   | n/a            | lost          |
| 3            | 0.126040    | 0.147714           | 0.148176         | 0.139155    | 0.138494   | 0.140268   | 0.000680   | n/a            | lost          |
| 4            | 0.057618    | 0.050900           | 0.050312         | 0.053507    | 0.052221   | 0.054724   | 0.000916   | n/a            | lost          |
| 5            | 0.021228    | 0.031464           | 0.030895         | 0.027057    | 0.025898   | 0.027900   | 0.000798   | 0.028157       | lost          |
| 6            | 0.067221    | 0.077425           | 0.076977         | 0.074051    | 0.072466   | 0.075588   | 0.001302   | n/a            | lost          |
| 7            | 0.035072    | 0.044793           | 0.044198         | 0.040907    | 0.040384   | 0.041585   | 0.000545   | n/a            | lost          |
| 8            | 0.073333    | 0.059363           | 0.058803         | 0.064814    | 0.063924   | 0.066330   | 0.000910   | n/a            | lost          |
| 9            | 0.366663    | 0.294754           | 0.298513         | 0.323584    | 0.319497   | 0.327765   | 0.003903   | n/a            | won           |
| 10           | 0.026021    | 0.024469           | 0.023943         | 0.025370    | 0.024802   | 0.025623   | 0.000324   | n/a            | lost          |
| 11           | 0.122221    | 0.152453           | 0.152996         | 0.142242    | 0.139446   | 0.144776   | 0.002270   | n/a            | lost          |
| 12           | 0.021228    | 0.026461           | 0.025920         | 0.023292    | 0.022920   | 0.023980   | 0.000411   | n/a            | lost          |

## SELECTED BETS

Race 2018062701, 2018, 12 runners.

| horse number | market prob | raw corrected prob | temperature prob | bagged prob | member min | member max | member std | estimated edge | actual result |
|--------------|-------------|--------------------|------------------|-------------|------------|------------|------------|----------------|---------------|
| 1            | 0.068158    | 0.049965           | 0.049587         | 0.059000    | 0.057533   | 0.059954   | 0.000924   | n/a            | lost          |
| 2            | 0.015432    | 0.013488           | 0.013225         | 0.014201    | 0.013886   | 0.014571   | 0.000293   | n/a            | lost          |
| 3            | 0.272631    | 0.265448           | 0.267536         | 0.269955    | 0.266133   | 0.271703   | 0.002232   | n/a            | won           |
| 4            | 0.148708    | 0.161397           | 0.161921         | 0.153768    | 0.149853   | 0.155852   | 0.002606   | n/a            | lost          |
| 5            | 0.022105    | 0.039990           | 0.039606         | 0.031385    | 0.030938   | 0.032303   | 0.000548   | 0.161261       | lost          |
| 6            | 0.043047    | 0.040466           | 0.040081         | 0.042176    | 0.041357   | 0.043551   | 0.000880   | n/a            | lost          |
| 7            | 0.062915    | 0.054896           | 0.054529         | 0.059280    | 0.058682   | 0.060054   | 0.000583   | n/a            | lost          |
| 8            | 0.122074    | 0.117780           | 0.117819         | 0.118775    | 0.117597   | 0.120000   | 0.001126   | n/a            | lost          |
| 9            | 0.048111    | 0.048587           | 0.048208         | 0.048109    | 0.047859   | 0.048439   | 0.000219   | n/a            | lost          |
| 10           | 0.048111    | 0.047751           | 0.047370         | 0.047576    | 0.047026   | 0.048087   | 0.000402   | n/a            | lost          |
| 11           | 0.123923    | 0.136251           | 0.136479         | 0.131607    | 0.130260   | 0.132928   | 0.001217   | n/a            | lost          |
| 12           | 0.024785    | 0.023981           | 0.023639         | 0.024168    | 0.024013   | 0.024333   | 0.000120   | n/a            | lost          |

# Betting and bankroll analysis

## Retrospective odds

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

## Do not select a threshold from these returns.

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

## SETTLEMENT BY STAKING RULE

| strategy                  | n bets | turnover  | net return    | roi       | total log growth | max drawdown | final bankroll | underwater days |
|---------------------------|--------|-----------|---------------|-----------|------------------|--------------|----------------|-----------------|
| flat_stake@edge>=0.02     | 174    | 1,740,000 | -973,500      | -0.559483 | -0.102420        | 0.098882     | 9,026,500      | 169             |
| kelly_no_rebate@0.01      | 338    | 110,870   | -578          | -0.005213 | -0.000058        | 0.002609     | 9,999,422      | 324             |
| kelly_settled_rebate@0.01 | 338    | 110,870   | -578          | -0.005213 | -0.000058        | 0.002609     | 9,999,422      | 324             |
| rebate_aware_kelly@0.01   | 339    | 155,100   | -10412.500000 | -0.067134 | -0.000742        | 0.004281     | 9992587.500000 | 325             |

Model view bagged\_offset. Seed -1. Every rule is settled on the same selected bets at the same odds, so the differences between these rows are stake sizing alone.

## EXPECTED AGAINST REALISED, BY ESTIMATED EDGE

### FLAT STAKE (EDGE>=0.02)

| bucket lower | bucket upper | n bets | mean estimated edge | expected return | realized return |
|--------------|--------------|--------|---------------------|-----------------|-----------------|
| 0.020000     | 0.050000     | 115    | 0.031990            | 0.031990        | -0.686957       |
| 0.050000     | 0.100000     | 50     | 0.069152            | 0.069152        | -0.516000       |
| 0.100000     | 0.200000     | 9      | 0.133063            | 0.133063        | 0.827778        |

### KELLY, NO REBATE (0.01)

| bucket lower | bucket upper | n bets | mean estimated edge | expected return | realized return |
|--------------|--------------|--------|---------------------|-----------------|-----------------|
| 0            | 0.020000     | 164    | 0.009823            | 0.009823        | 0.010487        |
| 0.020000     | 0.050000     | 115    | 0.031990            | 0.031990        | -0.445518       |
| 0.050000     | 0.100000     | 50     | 0.069152            | 0.069152        | -0.083108       |

| bucket lower | bucket upper | n bets | mean estimated edge | expected return | realized return |
|--------------|--------------|--------|---------------------|-----------------|-----------------|
| 0.100000     | 0.200000     | 9      | 0.133063            | 0.133063        | 3.606678        |

### KELLY, SETTLED REBATE (0.01)

| bucket lower | bucket upper | n bets | mean estimated edge | expected return | realized return |
|--------------|--------------|--------|---------------------|-----------------|-----------------|
| 0            | 0.020000     | 164    | 0.009823            | 0.009823        | 0.010487        |
| 0.020000     | 0.050000     | 115    | 0.031990            | 0.031990        | -0.445518       |
| 0.050000     | 0.100000     | 50     | 0.069152            | 0.069152        | -0.083108       |
| 0.100000     | 0.200000     | 9      | 0.133063            | 0.133063        | 3.606678        |

### REBATE-AWARE KELLY (0.01)

| bucket lower | bucket upper | n bets | mean estimated edge | expected return | realized return |
|--------------|--------------|--------|---------------------|-----------------|-----------------|
| unbounded    | 0            | 1      | -0.005611           | -0.005611       | 0.800000        |
| 0            | 0.020000     | 164    | 0.009823            | 0.009823        | -0.520066       |
| 0.020000     | 0.050000     | 115    | 0.031990            | 0.031990        | -0.445940       |
| 0.050000     | 0.100000     | 50     | 0.069152            | 0.069152        | 0.159636        |
| 0.100000     | 0.200000     | 9      | 0.133063            | 0.133063        | 3.606678        |

The check that matters most in this section: if the realised return in each edge bucket does not track the expected one, the edge estimate is wrong and every stake sized from it is wrong with it. The bucket counts here are small enough that one winner moves a bucket, so read the pattern across buckets rather than any single number.

### KELLY PRESETS

OPENING BANKROLL

HK\$10,000,000

PRESETS SETTLED

1% Kelly

PATH MODES

annual reset,  
continuous

SEEDS

5

7, 17, 42, 123, 256

BOOSTING ROUNDS

400

YEARS STAKED

2015-2026

Each path is rebased on its own opening balance, so zero means break-even for every line from the same point. Rebate policy: 10% of the race net loss, once that loss reaches HK\$10,000, as a cliff. Flat staking is omitted from these panels. Its stake is a fixed amount per bet and this settlement never halts on ruin, so its path can pass below zero -- a measure of how badly a rule loses rather than a balance an account could hold. It is also not a Kelly fraction, so it belongs to no preset. Its settlement is in the strategy table above. The bankroll config declares 2% Kelly, 5% Kelly, 10% Kelly as well, and this run did not settle them, so they have no button here: a preset that showed nothing would be worse than one that is absent.

1% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

+0.00284%

Kelly, no rebate, annual reset

BETS

338

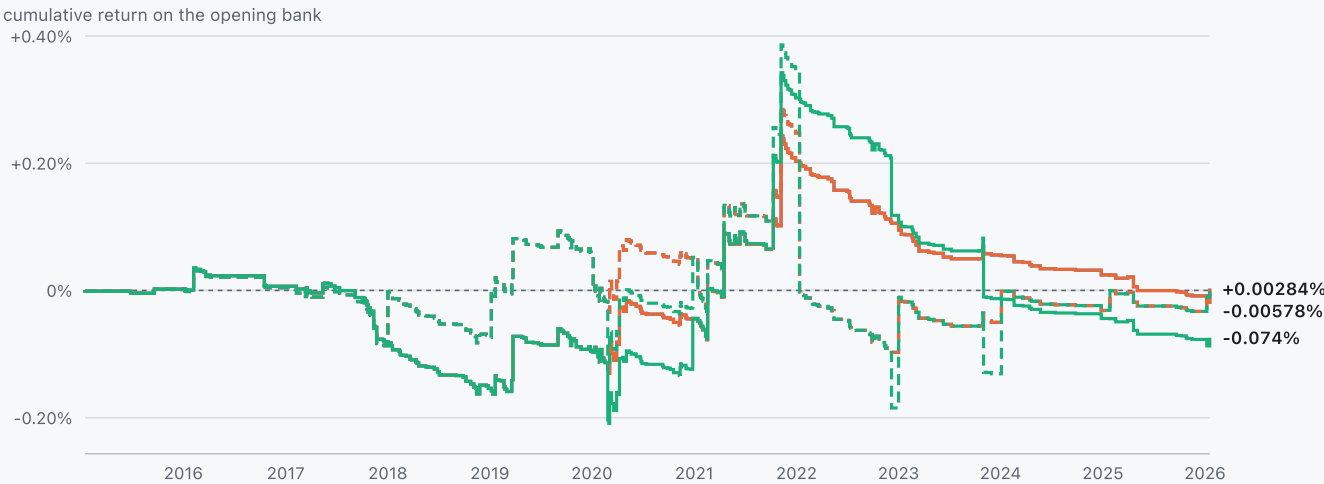
settled bets

REBATE RECEIVED

HK\$3,000

the most any rule received

Kelly, no rebate
Kelly, settled rebate
Rebate-aware Kelly



**Cumulative bankroll return at 1% Kelly** Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

| Series                              | Settlements | Final return | Note     |
|-------------------------------------|-------------|--------------|----------|
| Kelly, no rebate, annual reset      | 331         | +0.00284%    | 338 bets |
| Kelly, no rebate, continuous        | 331         | -0.00578%    | 338 bets |
| Kelly, settled rebate, annual reset | 331         | +0.00284%    | 338 bets |
| Kelly, settled rebate, continuous   | 331         | -0.00578%    | 338 bets |
| Rebate-aware Kelly, annual reset    | 332         | +0.00284%    | 339 bets |
| Rebate-aware Kelly, continuous      | 332         | -0.074%      | 339 bets |

BETS PER YEAR

| Year | Kelly, no rebate | Kelly, settled rebate | Rebate-aware Kelly |
|------|------------------|-----------------------|--------------------|
| 2015 | 3                | 3                     | 3                  |
| 2016 | 14               | 14                    | 14                 |
| 2017 | 37               | 37                    | 37                 |
| 2018 | 42               | 42                    | 42                 |



| Year | Kelly, no rebate | Kelly, settled rebate | Rebate-aware Kelly |
|------|------------------|-----------------------|--------------------|
| 2019 | 49               | 49                    | 49                 |
| 2020 | 58               | 58                    | 59                 |
| 2021 | 54               | 54                    | 54                 |
| 2022 | 37               | 37                    | 37                 |
| 2023 | 21               | 21                    | 21                 |
| 2024 | 10               | 10                    | 10                 |
| 2025 | 11               | 11                    | 11                 |
| 2026 | 2                | 2                     | 2                  |

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate and Kelly, settled rebate: 3 bets in 2015 against 11 in 2025. Rebate-aware Kelly: 3 bets in 2015 against 11 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Kelly, no rebate and Kelly, settled rebate (annual reset) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Kelly, no rebate and Kelly, settled rebate (continuous) settle identically, and none of them received any rebate: the threshold is a cliff, their stakes never reach it, so settling the rebate changes nothing. Rebate received: Rebate-aware Kelly, annual reset HK\$3,000; Rebate-aware Kelly, continuous HK\$3,000. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

### PRESET SENSITIVITY

| Preset   | Rule and mode                       | Opening        | Final          | Return    | Bets | Rebate    |
|----------|-------------------------------------|----------------|----------------|-----------|------|-----------|
| 1% Kelly | Kelly, no rebate, annual reset      | HK\$10,000,000 | HK\$10,000,284 | +0.00284% | 338  | HK\$0     |
| 1% Kelly | Kelly, no rebate, continuous        | HK\$10,000,000 | HK\$9,999,422  | -0.00578% | 338  | HK\$0     |
| 1% Kelly | Kelly, settled rebate, annual reset | HK\$10,000,000 | HK\$10,000,284 | +0.00284% | 338  | HK\$0     |
| 1% Kelly | Kelly, settled rebate, continuous   | HK\$10,000,000 | HK\$9,999,422  | -0.00578% | 338  | HK\$0     |
| 1% Kelly | Rebate-aware Kelly, annual reset    | HK\$10,000,000 | HK\$10,000,284 | +0.00284% | 339  | HK\$3,000 |
| 1% Kelly | Rebate-aware Kelly, continuous      | HK\$10,000,000 | HK\$9,992,588  | -0.074%   | 339  | HK\$3,000 |

The spread across presets is the point: a result that appears at only one Kelly fraction is a result about that fraction rather than about the model. A larger fraction stakes more of the bank on the same edge estimate, so it amplifies both the edge and the error in it.

### Simulation and uncertainty

### DAY-BLOCK BOOTSTRAP OF THE BANKROLL

| series                         | replicates | median terminal bankroll | terminal bankroll p5 | p below initial | median max drawdown | p95 max drawdown | p below min bet |
|--------------------------------|------------|--------------------------|----------------------|-----------------|---------------------|------------------|-----------------|
| flat_stake@edge>=0.02 / block1 | 2000       | HK\$9,011,645            | HK\$8,701,055        | 1.0000          | 0.1001              | 0.1304           | 0.0000          |
| flat_stake@edge>=0.02 / block3 | 2000       | HK\$9,027,154            | HK\$8,722,326        | 1.0000          | 0.0988              | 0.1292           | 0.0000          |
| flat_stake@edge>=0.02 / block7 | 2000       | HK\$9,022,940            | HK\$8,676,218        | 0.9995          | 0.0992              | 0.1339           | 0.0000          |



| series                                | replicates | median terminal<br>bankroll | terminal<br>bankroll p5 | p below<br>initial | median max<br>drawdown | p95 max<br>drawdown | p below<br>min bet |
|---------------------------------------|------------|-----------------------------|-------------------------|--------------------|------------------------|---------------------|--------------------|
| kelly_no_rebate@0.01 /<br>block1      | 2000       | HK\$9,998,828               | HK\$9,962,064           | 0.5175             | 0.0027                 | 0.0042              | 0.0000             |
| kelly_no_rebate@0.01 /<br>block3      | 2000       | HK\$9,997,755               | HK\$9,965,458           | 0.5385             | 0.0027                 | 0.0039              | 0.0000             |
| kelly_no_rebate@0.01 /<br>block7      | 2000       | HK\$9,998,701               | HK\$9,968,259           | 0.5265             | 0.0025                 | 0.0037              | 0.0000             |
| kelly_settled_rebate@0.01 /<br>block1 | 2000       | HK\$9,998,828               | HK\$9,962,064           | 0.5175             | 0.0027                 | 0.0042              | 0.0000             |
| kelly_settled_rebate@0.01 /<br>block3 | 2000       | HK\$9,997,755               | HK\$9,965,458           | 0.5385             | 0.0027                 | 0.0039              | 0.0000             |
| kelly_settled_rebate@0.01 /<br>block7 | 2000       | HK\$9,998,701               | HK\$9,968,259           | 0.5265             | 0.0025                 | 0.0037              | 0.0000             |
| rebate_aware_kelly@0.01 /<br>block1   | 2000       | HK\$9,992,109               | HK\$9,940,904           | 0.5870             | 0.0042                 | 0.0068              | 0.0000             |
| rebate_aware_kelly@0.01 /<br>block3   | 2000       | HK\$9,993,762               | HK\$9,947,417           | 0.5905             | 0.0040                 | 0.0065              | 0.0000             |
| rebate_aware_kelly@0.01 /<br>block7   | 2000       | HK\$9,992,879               | HK\$9,952,480           | 0.6110             | 0.0041                 | 0.0061              | 0.0000             |

Race days resampled with replacement, so results from one meeting stay together. This is sampling risk in the settled path, not model risk. 12 of 72 settled series shown, filtered to model view bagged\_offset.

## MODEL-ASSUMED MONTE CARLO

| series                    | replicates | median terminal<br>bankroll | terminal<br>bankroll p5 | p below<br>initial | median max<br>drawdown | p95 max<br>drawdown | p below<br>min bet |
|---------------------------|------------|-----------------------------|-------------------------|--------------------|------------------------|---------------------|--------------------|
| flat_stake@edge>=0.02     | 1000       | HK\$10,001,943              | HK\$9,203,076           | 0.4980             | 0.0514                 | 0.0920              | 0.0000             |
| kelly_no_rebate@0.01      | 1000       | HK\$10,003,246              | HK\$9,969,687           | 0.4390             | 0.0021                 | 0.0041              | 0.0000             |
| kelly_settled_rebate@0.01 | 1000       | HK\$10,003,246              | HK\$9,969,687           | 0.4390             | 0.0021                 | 0.0041              | 0.0000             |

Winners drawn from the model's own probabilities. 3 of 18 settled series shown, filtered to model view bagged\_offset.

### One simulation was declined rather than approximated

Refusals are recorded rather than omitted, so a reader can see the strategy was considered and declined. Rebate-aware sizing cannot reuse stale bankroll-dependent fractions on a divergent path.

The realised path, the day-block bootstrap and the model-assumed MC answer different questions and must not be shown as one uncertainty band.

### What a simulation cannot tell you

The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

CONFIGURATION

| Item                           | Value                                                                   |
|--------------------------------|-------------------------------------------------------------------------|
| Market offset column           | Implied_Prob                                                            |
| Features                       | 28                                                                      |
| Features the booster learns on | 27                                                                      |
| Opening bankroll               | 10,000,000                                                              |
| Bankroll mode                  | continuous                                                              |
| Rebate-aware sizing            | no                                                                      |
| Dataset registry digest        | sha256:f93edc409844f3df108c777ef6f48931b6b0d095d5b9b16fb1ae1a590da08844 |

definition, raw\_source, units, default\_behavior, higher\_value\_meaning and known\_caveats are intentionally blank. They are claims about the domain, not properties derivable from the model, and inventing them would produce documentation that reads as authoritative and is not.

KNOWN SOURCE-DATA DEFECTS

The lineage record carries {'landing\_dataset': 'aspx-results', 'landing\_duplicate\_keys': 1513, 'landing\_primary\_key': ['race\_id', 'Horse No.'], 'retained\_key': ['race\_id', 'Horse No.'], 'retained\_policy': 'keep\_first', 'retained\_policy\_source': 'features/build.py', 'feature\_table\_duplicate\_keys': 0, 'absorbed\_by\_build': 1513, 'consistent': True}. Recorded here rather than corrected silently: the feature build absorbs these, and a reader auditing a number should know they exist.

Limitations and the retrospective-odds warning

**Retrospective odds, restated**

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

### Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

### WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Bagged Offset (production) against the market. Beating a comparator and beating the market are different claims.
- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.

### Unverified source state

This run recorded a dirty source tree, so its numbers cannot be reproduced from the named commit alone.